Portfolio Risk Optimizer

Technical Reference Documentation

PortfolioRiskOptimizerApp.m — Complete Functional & Methodological Specification

MATLAB R2023b+ · App Designer · June 2026

1. Overview

1.1 Purpose

PortfolioRiskOptimizerApp is a professional-grade, multi-asset portfolio analytics workstation built as a MATLAB App Designer application. It is the primary tool for institutional portfolio risk measurement, strategic and tactical allocation, and quantitative stress analysis. All computation runs natively in MATLAB with no external runtime dependencies beyond the standard Optimization and Statistics & Machine Learning toolboxes.

The application covers the complete analytical workflow: loading or downloading market data; computing portfolio risk metrics across three VaR methodologies; optimising weights under eight objective functions; constructing and managing strategic (SAA) and tactical (TAA) allocations with full risk attribution; analysing correlation structure and principal components; tracing the efficient frontier; decomposing risk into per-asset and per-factor contributions; generating capital market assumptions (CMA); and running conditional block bootstrap stress tests.

1.2 Application Window

The application opens at 1560×980 pixels and is divided into a fixed **left control panel** (340 px wide, scrollable) and a **right analysis panel** that contains a 14-tab group. The left panel is always visible and provides all data loading, optimisation settings, and risk parameters. The right panel updates whenever Compute Risk or Optimize Weights is pressed.

Panel / Section	Controls	Function
Left — DATA	Load CSV, Sample Data, Download Yahoo, BBG Market, BBG Macro, Datastream, FRED Macro, Start/End dates, DataType, Freq, Custom Ticker, Project name, Save/Open Project	Load or download return/price series; set project metadata
Left — ASSETS	AssetsList (multi-select), CurrenciesList (Overlays)	Filter which funded assets and FX overlays are included in optimisation
Left — ACTIONS	Compute Risk, Optimize Weights, Export Weights CSV, Generate PDF Report	Trigger main computation or export pipelines
Left — OPTIMIZATION	Objective (8 choices), Optimizer (auto/quadprog/fmincon/ga), Rf rate, Target Return/Vol, Long-only checkbox, LB/UB bounds	Configure optimiser objective and weight constraints
Left — RISK	VaR Method (3 choices), Alpha (confidence level), Horizon (days)	Configure VaR/CVaR computation

Panel / Section	Controls	Function
Right — 14 tabs	Summary, Distribution, Sensitivity, Stress, Data, SAA, TAA, Combined, Perf, Bootstrap, Corr, Frontier, RiskDecomp, CMA	Full analytical output suite

1.3 Launch

```
cd C:\CIO\ClaudeCode\RiskToolpublic
run_PortfolioRiskOptimizerApp
```

The launcher clears the workspace and instantiates the app class. On startup the app reads the environment variable `FRED_API_KEY` if set, enabling immediate macro data download.

2. Data Sourcing

Five data sourcing paths are supported. All paths ultimately populate the internal state variables `app.R` (T×N return matrix), `app.AssetNames` (1×N cell array), and `app.Dates` (T×1 datetime vector).

2.1 Load CSV / Returns File

The **Load Returns / Prices CSV** button opens a file picker that accepts .csv and .txt files. The first column is interpreted as dates (any format that MATLAB's `readtable` can parse); remaining columns are asset series.

The **DataType** dropdown controls interpretation: *Prices* triggers simple-return computation (`(P_t / P_{t-1}) − 1`); *Returns* uses the values as-is. The **Freq** dropdown (Daily / Weekly / Monthly) sets the annualisation factor used throughout: 252, 52, or 12 respectively.

Custom Ticker: typing a ticker (e.g., `tlt.us`) into the Custom Ticker field and pressing **+ Add** downloads the series from Yahoo Finance / Stooq and appends it to the current dataset.

2.2 Sample Data

Pressing **Use Sample Data** generates a 750-observation (approximately 3 years daily) synthetic multi-asset return dataset using `generateRealisticSample`. The sample covers a representative mix of asset classes with realistic correlation structure and is designed for immediate exploration without any external data access. The sample is deterministically seeded (seed = 42) for reproducibility.

2.3 Yahoo Finance Download

The **Download Market Data** button downloads equity and ETF price series from the Yahoo Finance unofficial JSON API for the date range specified in the Start/End fields (format: YYYYMMDD). A configurable ETF map (`app.AssetETFMap`) associates asset names (e.g., "Equity_US") with Yahoo ticker symbols (e.g., "SPY"). Data is fetched via webread with a standard browser User-Agent header to avoid 429 rate-limit rejections. Results are converted to daily simple returns and stored in the same format as CSV data.

2.4 Bloomberg

Two Bloomberg buttons are provided, both requiring an active Bloomberg terminal session and the MATLAB Datafeed Toolbox:

- **BBG Market**: downloads daily price history for all assets in the ETF map using the `history()` function. Overrides request calendar-day fill with previous-value carry for non-trading days.

- **BBG Macro**: downloads macro series (GDP, CPI, PMI, etc.) for the same date range and stores them as measures available to the Bootstrap condition builder.

Bloomberg API note	Newer BLPAPI requires date arguments as 'mm/dd/yyyy' char strings (not MATLAB datenums) and overrides as an N×2 cell array. Both are handled internally. If BBG is unavailable, Yahoo Finance is a drop-in substitute for most equity and ETF series.

2.5 FRED Macro Download

The **Download Macro (FRED)** button fetches twelve macroeconomic series from the Federal Reserve FRED REST API, aligns them to the loaded return dates, and stores the result in `app.Measures` for use in the Bootstrap condition builder. Available series include: GDP growth, CPI, nonfarm payrolls, retail sales, VIX, Fed Funds rate, 2Y and 10Y Treasury yields, yield curve slope, HY and IG credit spreads, oil price (WTI), and USD momentum (DXY). A free API key from fred.stlouisfed.org is required and read from the `FRED_API_KEY` environment variable.

2.6 Project Persistence

The **Save Project** button serialises the full application state — returns matrix, asset names, weights, SAA and TAA tables, all settings — to a `.mat` file. **Open Project** restores this state exactly, including all tab displays. Project files allow sessions to be archived, shared, and audited without re-downloading data.

3. Summary Tab — Core Risk Metrics

The Summary tab is the primary output panel. It is refreshed every time **Compute Risk** is pressed, or whenever data is loaded or weights change. It contains two components: a text area with all portfolio-level metrics, and a table showing per-asset weights and risk contributions.

3.1 Portfolio Return Series

The portfolio return at time t is the weighted sum of asset returns:

```
rp = R * w;   % T×1 portfolio return vector
```

where `R` is the T×N return matrix and `w` is the N×1 weight vector. The current weights are stored in `app.W` and reflect the combined SAA+TAA allocation after any TAA overlay is applied.

3.2 Return and Volatility

Metric	Per-Period Formula	Annualised
Mean Return	μ_p = w′μ (where μ = mean(R, axis=0))	μ_ann = μ_p · k
Volatility	σ_p = √(w′Σw) (where Σ = cov(R))	σ_ann = σ_p · √k
Sharpe Ratio	(μ_p – r_f) / σ_p	(μ_ann – r_f·k) / σ_ann
Sortino Ratio	(μ_p – r_f) / σ_down	(μ_ann – r_f·k) / (σ_down · √k)
Calmar Ratio	μ_ann / MaxDD	—
Max Drawdown	max(CumMax_t – CW_t) / CumMax_t	—

Downside deviation σ_down is the root-mean-square of negative excess returns only: σ_down = `sqrt(mean(r_excess[r_excess < 0].^2))`. The annualisation factor k is 252 (daily), 52 (weekly), or 12 (monthly).

3.3 Value at Risk Methods

VaR is reported as a positive number representing the loss at confidence level α over a horizon of h periods. Three methods are implemented:

Historical Simulation.

The (1–α) empirical quantile of the negative of the portfolio return distribution. No distributional assumption is made. Naturally captures fat tails, skewness, and any non-linearity present in actual returns. Horizon scaling is not applied to historical VaR/CVaR — each observation already represents one full period.

```
VaR_hist = -quantile(rp, 1-alpha);
```

Parametric (Normal).

Assumes portfolio returns follow $N(\mu_p, \sigma^2_p)$. VaR is computed analytically using the standard normal quantile z_α. The horizon is incorporated via the square-root-of-time scaling, which is exact under i.i.d. normality:

```
z_alpha = norminv(1-alpha);
VaR_param  = -(mu_p * h + z_alpha * sig_p * sqrt(h));
CVaR_param = -(mu_p * h - sig_p * sqrt(h) * normpdf(z_alpha) / (1-alpha));
```

Monte Carlo (Normal).

Draws 20,000 simulated portfolio returns from $N(\mu_p \cdot h, \sigma^2_p \cdot h)$. VaR and CVaR are computed as empirical quantiles of this simulated distribution. Uses the fixed random seed (default 42) for reproducibility. This method allows for non-parametric tail estimation from a normal generator, which is useful for verifying the parametric formulas or extending to non-normal generators in future.

```
rng(app.RandSeed);
simRet = mu_p*h + sig_p*sqrt(h)*randn(MCPaths,1);
VaR_mc  = -quantile(simRet, 1-alpha);
CVaR_mc = mean(-simRet(simRet < -VaR_mc));
```

3.4 Multi-Level Expected Shortfall

The Summary panel additionally reports historical VaR and CVaR at three fixed confidence levels simultaneously: 90%, 95%, and 99%. This provides a quick tail-risk profile without requiring the user to rerun the computation at different α values. All three are computed using historical simulation regardless of the selected VaR method.

3.5 Higher Moments

Skewness (third standardised moment) and excess kurtosis (fourth moment minus 3) are reported for the portfolio return series. Negative skewness is common in equity portfolios and indicates a longer left tail than right tail. Positive excess kurtosis (leptokurtosis) indicates fatter tails than the normal distribution — a key risk not captured by variance-based measures alone.

3.6 Per-Asset Risk Contributions

The WeightsTable in the lower half of the Summary tab shows the per-asset Euler (marginal) risk contribution to portfolio variance:

```
MCR = (Sigma * w) / sigma_p;     % marginal contribution to vol
RC  = w .* MCR;                  % component risk contribution (vol units)
```

The sum of all component contributions equals the portfolio volatility: $\Sigma RC_i = \sigma_p$. $RC_i > 0$ for risk-adding positions and $RC_i < 0$ for risk-reducing (short) positions.

4. Distribution Tab

The Distribution tab displays the empirical portfolio return distribution as a histogram with a kernel density estimate (KDE) overlay. Vertical dashed lines mark the VaR and CVaR thresholds at the configured confidence level, with the tail region shaded to visually communicate the severity of extreme losses. The chart updates with every Compute Risk call and uses the currently selected VaR method to position the markers.

Interpretation	The gap between the VaR marker (the quantile itself) and the CVaR marker (the average of losses beyond VaR) is a visual measure of tail thickness. A wide gap indicates a fat left tail and suggests that VaR significantly understates potential extreme losses.

5. Sensitivity Tab

The Sensitivity tab provides two complementary views of portfolio sensitivity to small moves.

5.1 Asset Shock Sensitivity

The upper chart shows the delta P&L impact of a +1% move in each individual asset, holding all other assets constant. This is a first-order linear approximation:

```
dPnL_i = w_i * 0.01;   % for each asset i
```

Bars are sorted by absolute magnitude. Long positions produce positive delta PnL (rising asset helps portfolio); short or underweight positions produce negative or zero delta PnL. This chart quickly identifies which assets drive the most directional P&L sensitivity.

5.2 Factor Sensitivity

The lower chart shows the portfolio P&L impact of a +1% move in each factor, where factors are defined by the asset-to-factor mapping:

Asset Name	Factor Label
Equity_US	Equity
HG_Bonds_7Y	Rates
HY_Bonds	Credit
EUR_USD	FX
Commodities	Commodities
Gold	Gold

Factor sensitivities are the sum of portfolio weights for all assets mapped to that factor, multiplied by 0.01. This view is useful for communicating directional risk in terms of familiar market factors rather than individual instruments.

6. Stress Tab — Scenario Analysis

The Stress tab allows the user to define and evaluate named stress scenarios against the current portfolio. Each scenario specifies a set of per-asset returns (shocks), and the tab computes the resulting portfolio P&L as a linear approximation.

6.1 Scenario Table

The ScenTable is fully editable. Returns are entered as decimals (e.g., –0.05 = –5%). A scenario must include a return for each asset in the portfolio; assets not explicitly shocked receive a return of 0. The portfolio scenario P&L is computed as:

```
PnL_scenario = w' * asset_shocks;
```

6.2 Pre-Built Historical Stress Episodes

The **Load All Stresses** button populates the table with six canonical historical stress episodes based on realised maximum drawdowns during each event:

Episode	Approximate Period	Characteristic
GFC (Oct 2008)	2008-09 to 2009-03	Systemic deleveraging; equity –55%, credit spreads +2000bp
COVID (Mar 2020)	2020-02 to 2020-03	Fastest 30% equity drawdown on record; cross-asset correlation spike
Rate Shock (2022)	2022-01 to 2022-06	Simultaneous bond and equity losses; worst 60/40 year since 1970
EM Crisis (1998)	1998-08 to 1998-10	Russian default + LTCM; EM and HY spreads surge
Inflation Spike	Custom	Elevated inflation shock: real assets up, nominal bonds down
USD Surge	Custom	Strong dollar episode: USD-denominated assets rise, EM and commodities decline

Alternatively, the **Scenario** dropdown lets the user select any single pre-built episode and view only that shock. The ScenStatsText panel below the table shows total P&L, approximate dollar impact, and rank among all loaded scenarios.

7. SAA Tab — Strategic Asset Allocation

The SAA tab implements the first layer of a two-layer portfolio construction process. The Strategic Asset Allocation (SAA) represents the long-run target portfolio — the neutral position from which tactical deviations are measured.

7.1 Funded Allocations

The **Funded Allocations** table (SAATable) shows assets that are subject to the full-investment constraint (weights sum to 100%). Columns are:

Column	Editable	Description
Asset	No	Asset name from the loaded dataset
WeightPct	Yes	Weight as a percentage (e.g., 40.0 = 40%)
Active	Yes	Checkbox — whether this asset is included in optimisation
Description	Yes	Free-text label (e.g., "US Large Cap Equity", "7Y Investment Grade Bonds")

Weights are editable inline. The **Normalize Combined to 100%** checkbox (when ticked) automatically rescales funded weights to sum to 100% whenever the TAA overlay is applied. The **Reset SAA to Default** button resets all funded weights to equal-weight across all active assets.

7.2 Overlay Positions

The **Overlay Positions** table (SAAFXTable) shows instruments that do not participate in the full-investment constraint — typically FX forwards, overlay strategies, and currency hedges. These are identified by an internal `isFXOverlay` heuristic that flags any asset whose name contains "USD", "EUR", "GBP", "JPY", "FX", "forward", "overlay", or "hedge".

The FXExposureText shows a one-line summary of gross and net FX exposure derived from the overlay table: `Gross FX = Σ|w_overlay|`, `Net FX = Σw_overlay`.

7.3 SAA Risk/Return Analytics

The SAAMetricsText panel computes and displays the full risk-return profile for the SAA portfolio in isolation, using the same metric set as the Summary tab: annualised return, volatility, Sharpe ratio, Sortino ratio, Calmar ratio, max drawdown, VaR, and CVaR. This allows comparison of the SAA profile against the current combined (SAA+TAA) portfolio displayed in the Summary tab.

7.4 SAA Stress Scenario Performance

The SAAStressTable shows P&L for the six canonical stress episodes applied to the SAA weights specifically. This is distinct from the Stress tab, which shows the combined portfolio. Comparing

SAA stress results against the Combined stress results (visible in the Stress tab) quantifies how much the TAA overlay adds or subtracts from stress resilience.

8. TAA Tab — Tactical Asset Allocation

The TAA tab implements the second layer of the two-layer allocation framework. Tactical positions are expressed as delta deviations on top of the SAA weights. A positive delta means overweight vs SAA; a negative delta means underweight.

8.1 Building the TAA Book

Two methods are available for adding positions to the TAA table:

- **Add Selected:** Select one or more assets from the asset list above and press Add Selected. Each selected asset appears as a new row in the TAATable with DeltaPct = 0. Edit the DeltaPct column inline to set the desired delta.
- **Add Trade Pair:** Select exactly two assets and press Add Trade Pair. A paired trade is created: the first asset gets DeltaPct = +1.0 (Long) and the second gets DeltaPct = –1.0 (Short). Both legs are grouped under the same TradeGroup name (e.g., "Trade1", "Trade2"). This facilitates relative-value trade entry.

8.2 TAA Table Columns

Column	Type	Editable	Description
TradeGroup	string	Yes	Groups related legs. Assigned automatically ("Trade1", "Trade2", …) but editable.
Asset	string	Yes	Asset ticker / name matching the SAA universe.
DeltaPct	double	Yes	Weight delta in percent (e.g., 5.0 = +5%). Can be negative.
Leg	string	No	Auto-derived: "Long" if DeltaPct > 0, "Short" if DeltaPct < 0, "Flat" if zero.
Description	string	Yes	Optional free-text trade rationale or label.
Standalone_TE	double	No	Tracking error of this position if it were the only active deviation (bps, annualised). Computed on Apply.
Marginal_TE	double	No	Incremental TE contribution: TE_all minus TE_without_this_position (bps). Computed on Apply.
DivBenefit	double	No	Standalone_TE minus Marginal_TE (bps). Positive = diversifying vs rest of book.
EqBeta	double	No	Equity beta: Σ(asset, Equity_US) / Σ(Equity_US, Equity_US). Measures equity sensitivity.
USD_Sens	double	No	USD sensitivity: Σ(asset, USD_Cash) / Σ(USD_Cash, USD_Cash). Measures dollar sensitivity.

8.3 Trade-Level Summary Table

Below the main TAATable, the **TAATradeTable** aggregates by TradeGroup. For each trade group, it shows the Standalone_TE, Marginal_TE, and DivBenefit of the combined trade (both legs together, not summing individual legs), plus the weighted-average EqBeta and USD_Sens. The trade-level TE is computed by treating the entire group's active weight vector as a single isolated bet, exactly as for individual rows but with all legs of the trade included.

The header above TAATradeTable reads: *"Trade-Level Summary (all TE in bps ann.)"*

8.4 Applying the TAA to the Combined Portfolio

The **Apply TAA** → **Combined** button (highlighted green) merges the SAA weights with the TAA deltas. The combined weight for each funded asset is:

```
w_combined(i) = w_SAA(i) + DeltaPct(i)/100;  % DeltaPct is in %

% If NormalizeCheck is ticked:
w_combined = w_combined / sum(w_combined);    % re-normalise funded assets to 100%
```

Overlay (FX) weights are preserved as-is and do not participate in the normalisation. After Apply TAA, all dependent tabs (Combined, Perf, Bootstrap, RiskDecomp, Corr) are refreshed automatically. The TAATable TE columns are populated at this point using the selected TE method.

9. Combined Tab

The Combined tab shows the final merged portfolio (SAA + TAA) and provides tracking error analysis relative to the SAA benchmark.

9.1 Funded Positions & Overlay Positions

Two tables mirror the SAA tab structure: CombinedTable shows funded assets (Asset, Weight, RiskContribution), and CombFXTable shows overlay positions with the same columns. Both reflect the post-Apply-TAA combined weights. The CombAnalyticsText panel below displays summary analytics:

- **Information Ratio**: (ann. active return) / (ann. tracking error)
- **Hit Rate**: fraction of periods where active return ($r_p - r_{bm}$) > 0
- **ETL**: Expected Tail Loss (CVaR) of active returns
- **CDaR**: Conditional Drawdown at Risk — average of drawdowns exceeding the α threshold

9.2 Tracking Error Methodology

The **TEMethodDropDown** control (top-right of the Combined tab) selects the covariance estimation method used for all TE computations in this tab and in the TAATable columns. Three options are available:

Var-Covar (default).

Uses the full-history sample covariance matrix. Provides a stable, unbiased estimate suitable for long-horizon allocations. $TE = \sqrt{(w_{diff}' \Sigma w_{diff})} \cdot \sqrt{k}$, where $w_{diff} = w_{TAA} - w_{SAA}$.

Historical.

Computes realised active returns $r_t = (R \cdot w_{TAA} - R \cdot w_{SAA})$ for each period and reports the annualised standard deviation of this series. This is a backward-looking, model-free TE estimate that captures actual realised tracking error including regime changes. $TE = \mathrm{std}(R \cdot w_{diff}) \cdot \sqrt{k}$.

Exp Weighted ($\lambda = 0.94$).

Applies exponentially declining weights to the covariance matrix, giving more weight to recent observations. Uses the RiskMetrics standard decay factor $\lambda = 0.94$. The weighted covariance is:

```
[T, N] = size(R);
w_t = 0.94 .^ (T-1:-1:0)';          % exponential weights (oldest first)
w_t = w_t / sum(w_t);                % normalise to sum to 1
mu  = w_t' * R;                      % weighted mean
Rc  = R - repmat(mu, T, 1);          % demeaned returns
S   = Rc' * diag(w_t) * Rc;         % weighted outer product
S   = (S + S') / 2;                 % symmetrise
```

```
TE  = sqrt(w_diff' * S * w_diff) * sqrt(k);
```

9.3 Tracking Error Attribution Table

The TETable (bottom of Combined tab) shows per-asset TE attribution using the Euler decomposition consistent with the selected TE method:

Column	Formula	Units
Asset	—	—
ActiveWt %	w_diff_i × 100	percent
TE_RC bps	w_diff_i · (Σ w_diff)_i / √(w_diff′Σw_diff) · √k · 10000	basis points per annum
Share %	TE_RC_bps_i / Σ TE_RC_bps_j × 100	percent of total TE

Sum of TE_RC_bps across all assets equals the total portfolio tracking error in bps. Share % summing to 100% confirms the decomposition is exact. Negative TE_RC values indicate risk-reducing active positions (the asset is negatively correlated with the rest of the active book).

10. Portfolio Optimisation

Pressing **Optimize Weights** solves a constrained optimisation problem for the funded assets. Overlay assets are excluded from optimisation and their weights are preserved. Eight objective functions are available.

10.1 Constraints

All objectives share the following constraint structure:

Constraint	Formula	Default
Full investment (funded)	$\Sigma w_i = 1$ (funded assets only)	Always enforced
Long-only	$w_i \geq 0$ for all i	Enabled (Long-only checkbox)
Lower bound	$w_i \geq LB$	LB = 0
Upper bound	$w_i \leq UB$	UB = 1
Target return	$\mu'w$ = TargetRet (if set)	Not set by default

10.2 Objective Functions

Objective	Mathematical Problem	Solver
Min Variance	min $w'\Sigma w$ subject to: full-investment, bounds, optional target return	quadprog (preferred) or fmincon SQP
Max Sharpe	max $(\mu'w - r_f) / \sqrt{(w'\Sigma w)}$ subject to: full-investment, bounds	fmincon SQP
Equal Weight	$w_i = 1/N$ for all selected funded assets	Analytical (no solver)
Target Vol	Step 1: find tangency portfolio (Max Sharpe) Step 2: scale tangency portfolio to achieve σ_{target} = TargetRet field Excess weight goes to / from cash	fmincon SQP (Step 1)
Min CVaR (hist)	min $CVaR_\alpha(R \cdot w)$ where $CVaR_\alpha$ = mean of returns below VaR_α threshold Optional target return constraint	fmincon SQP (or GA if selected)
Min VaR (hist)	min $VaR_\alpha(R \cdot w) = -\text{quantile}(R \cdot w, 1-\alpha)$	fmincon SQP (or GA if selected)
Risk Parity (ERC)	min $\Sigma_{ij} (RC_i - RC_j)^2$ where $RC_i = w_i \cdot (\Sigma w)_i$ Cash assets forced to w=0	fmincon SQP, initialised from inverse-vol weights
Black-Litterman	Blend equilibrium returns $\pi_{eq} = \delta \cdot \Sigma \cdot w_{mkt}$ with TAA views P, Q via BL posterior: $\mu_{BL} = \pi + M \cdot (Q - P \cdot \pi)$ Max Sharpe on μ_{BL} (risk aversion δ=2.5, τ=0.05)	fmincon SQP

10.3 Solver Selection Logic

The **Optimizer** dropdown controls the back-end solver: `auto` (default), `quadprog`, `fmincon`, or `ga`. In auto mode, the selection logic is:

```
if objective == 'Min Variance' && quadprog available:
    use quadprog     % exact QP, fastest
elif fmincon available:
    use fmincon      % SQP, handles all nonlinear objectives
elif ga available:
    use ga           % genetic algorithm, last resort for non-smooth objectives
else:
    use fmincon      % always available as Optimization Toolbox is required
```

10.4 Black-Litterman Implementation

The BL model uses the current SAA weights as the market-cap proxy w_mkt. The equilibrium implied excess return vector is computed as π_eq = $\delta \cdot \Sigma \cdot$w_mkt with risk-aversion $\delta = 2.5$. Views P and Q are built from the current TAA table: each asset with a non-zero DeltaPct contributes a view that the asset will outperform or underperform the market-implied return in proportion to the delta. The posterior mean is:

```
Omega = diag(diag(P * (tau*Sigma) * P'));  % view uncertainty (diagonal)
M     = tau*Sigma*P' * inv(P*tau*Sigma*P' + Omega);
mu_BL = pi_eq + M * (Q - P*pi_eq);
```

A Max Sharpe optimisation is then run using μ_BL in place of the historical mean. If no TAA views exist, the BL result is identical to standard Max Sharpe with equilibrium-implied returns.

11. Performance Tab

The Performance tab provides rolling and cumulative analytics for the current combined portfolio. It contains three time-series charts and one summary table.

Chart / Control	Content	Update Trigger
Cumulative Performance (PerfAxesCum)	Cumulative compound return: cumprod(1 + rp) – 1. Plotted as percentage.	Compute Risk, data load, weight change
Rolling Volatility (PerfAxesVol)	Annualised rolling volatility over a configurable window (default 63 days). Uses std(rp_window) · √k.	Compute Risk or window change
Drawdown (PerfAxesDD)	Underwater equity chart: (CW_t – CumMax_t) / CumMax_t. Shows all drawdown episodes.	Compute Risk, data load
Performance Table (PerfTable)	Per-sub-period breakdown: rolling window Sharpe, vol, return, Sortino, max DD, hit rate. Rows correspond to non-overlapping rolling windows.	Rolling window field change
Rolling Window field	Number of periods for rolling calculations (default 63). Min 2.	User input

Rolling volatility interpretation	A 63-period (approximately 3-month) window is the default. Shorter windows respond more quickly to volatility regime shifts; longer windows are more stable. Compare the rolling vol chart against the cumulative return chart to identify periods where volatility was elevated — these typically correspond to drawdown episodes.

12. Bootstrap Tab — Conditional Stress Testing

The Bootstrap tab implements a conditional historical block bootstrap for stress scenario analysis. Unlike simple Monte Carlo, this method preserves the empirical autocorrelation and cross-sectional dependency structure present in real financial return data — properties that are especially important during stress episodes.

12.1 Methodology

A **block** is a contiguous segment of L consecutive trading-day returns drawn at random from the historical return matrix. Concatenating B randomly selected blocks produces one simulated path of approximately B·L observations. The **conditional** aspect means that only historical blocks beginning on dates that satisfy all (or any) user-specified regime conditions are eligible for sampling — effectively filtering the bootstrap pool to a specific macro or market environment.

Portfolio P&L is computed along each simulated path by applying the current combined portfolio weights. The resulting distribution of path-level annualised returns, VaR, CVaR, and maximum drawdown are then summarised statistically.

12.2 Condition Builder

The BootCondTable defines the sampling filter as a set of rules. Each row specifies:

Column	Options	Description
Measure	GDP Growth (YoY), Inflation (CPI YoY/MoM), Nonfarm Payrolls (MoM), Retail Sales (YoY), USD Momentum (DXY 3m), VIX Level, Fed Funds Rate, 2Y/10Y Treasury Yield, Yield Curve Slope (10Y-2Y), HY/IG Credit Spread, Oil Price (WTI 3m)	The macro or market variable to test
Condition	> above, < below, >= at or above, <= at or below	Comparison operator
Value	Numeric threshold	Threshold in the native units of the measure (e.g., 0.02 = 2% for GDP growth)

Multiple conditions are combined using the **BootCondLogicDrop**: *AND (all must hold)* restricts sampling to periods where every rule is simultaneously satisfied; *OR (any must hold)* includes any period where at least one rule is satisfied. The default condition set is `GDP_YoY > 0.02 AND DXY_3m < 0` (growth environment with a weakening dollar).

Measures are loaded either from the **Load Measures CSV** button (expects a date-indexed CSV) or from the **Download Macro (FRED)** button on the left panel.

12.3 Configuration Parameters

Parameter	Field	Default	Description
Block Length	BlockLenField	20 days	Length of each bootstrap block in trading days. Longer blocks preserve more autocorrelation; shorter blocks increase path diversity.
Paths	PathsField	500	Number of simulated paths. More paths improve distributional accuracy. Typically 500–2000 is sufficient.
Horizon	From VaR tab	1 period	Risk horizon. Determines how many blocks to concatenate per path: ceil(Horizon / BlockLen).
Condition logic	BootCondLogicDrop	AND	AND: all conditions must hold simultaneously. OR: any single condition suffices.

12.4 Output

After pressing **Run Bootstrap**, the tab displays three outputs:

- **BootstrapAxes**: histogram of annualised path returns across all simulated paths, with vertical lines for the mean, 5th percentile (downside VaR), and 95th percentile.
- **BootCondAxes**: a timeline chart showing which historical dates matched the condition set (shaded), helping the user assess how restrictive the filter is and whether the resulting sample is large enough to be statistically meaningful.
- **BootStatsTable**: statistics of the simulated return distribution: mean, median, std, 5th/25th/75th/95th percentiles, VaR, CVaR, and expected max drawdown.
- **BootstrapSummary text**: one-paragraph narrative summary including the number of eligible dates, number of paths run, and key tail statistics.

13. Correlation Tab

The Correlation tab provides two views of pairwise return correlations and a principal component decomposition of the covariance matrix.

13.1 Correlation Heatmap

The CorrAxes panel renders the full N×N correlation matrix as a colour-coded heatmap. Cells are coloured on a diverging scale from –1 (red) to +1 (blue), with white at zero. Asset names are labelled on both axes. The matrix is computed from the full return history using the sample correlation estimator.

13.2 Rolling Correlation

The CorrRollAxes panel shows the rolling correlation between each individual asset's return and the portfolio return, over a configurable window (default 63 periods). A correlation of +1.0 means the asset moves perfectly with the portfolio; 0 means no linear relationship; –1.0 means it perfectly offsets the portfolio. This chart helps identify which assets provide the most consistent diversification over time versus which are highly correlated with the portfolio. The window is adjustable via the **Rolling window** field, which triggers an immediate redraw.

13.3 PCA Eigenvalue Table

The EigenTable shows the principal component decomposition of the correlation matrix. Columns are: PC (principal component number), Eigenvalue, PctVariance (% of total variance explained by this PC), and CumulativePct (running sum). The number of PCs needed to explain 90%+ of variance is a measure of the effective dimensionality of the portfolio — a low number indicates high correlation clustering; a high number indicates more independent sources of return.

```
[V, D] = eig(corr(R), 'vector');
[lambda, idx] = sort(D, 'descend');
pct_var = lambda / sum(lambda) * 100;
```

14. Efficient Frontier Tab

The Frontier tab traces the mean-variance efficient frontier and locates the current portfolio on it.

14.1 Frontier Computation

Pressing **Compute Frontier** solves the parametric quadratic programme for a grid of target returns spanning the range from the minimum-variance portfolio return to the maximum achievable return. The number of points on the grid is configurable via the **Frontier points** field (default 50, min 10, max 500). At each target return level, the optimiser solves:

```
% For each target return mu_t in the grid:
min  w' * Sigma * w
s.t. sum(w) = 1,  w' * mu = mu_t,  LB <= w <= UB
```

Only funded assets are included in the frontier. Overlay weights are excluded but their presence in the portfolio does not affect the frontier visualisation, which shows funded-asset-only efficient portfolios.

14.2 Frontier Chart

The FrontierAxes plots annualised volatility (x-axis) against annualised return (y-axis) for all computed efficient portfolios. Three special portfolios are highlighted:

- **Minimum Variance Portfolio**: the leftmost point on the frontier (lowest vol).
- **Maximum Sharpe Portfolio**: the tangent portfolio (highest Sharpe ratio), shown with a star.
- **Current Portfolio**: plotted as a circle with its exact coordinates.

Individual assets are also plotted as crosses on the same chart, showing their stand-alone risk-return characteristics and making diversification benefits visually apparent (the portfolio frontier lies to the left of all individual assets).

14.3 Distance to Frontier

The **FrontierDistLabel** below the chart reports the Euclidean distance in (vol, return) space from the current portfolio to the nearest point on the efficient frontier. A non-zero distance means the current allocation is mean-variance inefficient — it could achieve either the same return with less risk, or more return with the same risk, by rebalancing along the frontier.

15. Risk Decomposition Tab

The RiskDecomp tab provides a comprehensive decomposition of portfolio risk into per-asset and per-factor contributions, concentration metrics, and rolling factor exposure analysis.

15.1 Risk Budget Table

The RiskBudgetTable (top-left quadrant) shows the Euler-based risk decomposition for each asset, split by allocation source (SAA vs TAA):

Column	Formula	Units
Asset	—	—
Source	"SAA", "TAA", or "SAA+TAA"	—
SAA_Wt	w_SAA_i × 100	%
TAA_Wt	w_TAA_i × 100	%
TotalWt	(w_SAA_i + w_TAA_i) × 100	%
MCR_bps	(Σw)_i / σ_p × √k × 10000	basis points per annum
CR_SAA_bps	w_SAA_i × MCR_i × √k × 10000	bps (SAA portion only)
CR_TAA_bps	w_TAA_i × MCR_i × √k × 10000	bps (TAA portion only)
CR_Total_bps	(w_SAA_i + w_TAA_i) × MCR_i × √k × 10000	bps
PctRisk	CR_i / σ_p × 100	%

MCR (Marginal Contribution to Risk) is the rate of change of portfolio volatility with respect to a marginal increase in asset i's weight. It is computed as (Σw)_i / σ_p, where (Σw)_i is the i-th element of the Sigma·w vector. MCR is the same regardless of whether the weight is SAA or TAA — it reflects the current combined portfolio structure.

15.2 Risk Waterfall Chart

The RiskWaterfallAxes (top-right) renders a horizontal bar chart of per-asset total risk contributions sorted in descending order. Positive contributions (long positions adding risk) are shown in red; negative contributions (short or strongly negatively-correlated positions) are shown in green. The chart title shows the total portfolio annualised volatility in basis points. Percentage-of-total labels are printed on each bar for immediate readability.

15.3 Concentration Metrics

The ConcentrationText panel reports four concentration metrics:

Metric	Formula	Interpretation
HHI (weights)	$\Sigma(\|w_i\| / \Sigma\|w_j\|)^2$	Weight concentration: 1/N for equal weight, 1.0 for single-asset
HHI (risk)	$\Sigma(\|RC_i\| / \Sigma\|RC_j\|)^2$	Risk concentration: 1/N for equal risk contribution, 1.0 for single-driver
Effective Bets (ENB)	1 / HHI(risk)	Effective number of bets: N for equal risk, 1 for single-driver
Diversification Ratio	$\Sigma\|w_i\|\cdot\sigma_i / \sigma_p$	Ratio of weighted average vol to portfolio vol. Higher = more diversification benefit

15.4 Factor Decomposition

The FactorExposureTable decomposes portfolio risk into factor and specific (idiosyncratic) components via OLS regression of portfolio returns on factor returns:

```
% OLS: r_p = alpha + beta_1*F_1 + ... + beta_k*F_k + epsilon
X = [ones(T,1), R(:, factorIdx)];
beta = (X'*X) \ (X'*r_p);
residuals = r_p - X*beta;
R2 = 1 - var(residuals)/var(r_p);
```

Columns shown: Factor name, Beta, t-statistic (beta / SE), and Contribution ($\beta_i \times cov(F_i, r_p) / var(r_p) \times 100\%$). The FactorVsSpecificText panel below the table reports the factor-explained fraction of total variance (R^2) and the residual specific variance, annualised in bps.

The RollingFactorAxes chart shows rolling betas over a configurable window (default 126 periods, set via FactorWindowField). Stable rolling betas indicate a consistent factor exposure; time-varying betas may reveal regime shifts or structural changes in the portfolio's factor composition.

16. CMA Tab — Capital Market Assumptions

The CMA tab generates forward-looking expected returns and volatility for each asset using three methodologies. The output can be used directly as optimiser inputs by pressing **Use CMA as Optimizer Inputs**.

16.1 CMA Methods

Historical.

Computes sample mean and standard deviation from the loaded return history. A configurable lookback (in periods; 0 = use full history) limits the sample window. Simple and transparent but can be heavily influenced by the historical period selected, including start/end regime effects.

Building-Block.

Constructs expected returns by adding a risk premium over the risk-free rate, where the risk premium is proportional to the asset's historical Sharpe ratio scaled by a target market Sharpe. This method is less sensitive to the historical period than the pure historical approach and allows the analyst to express a view on the equilibrium Sharpe ratio.

Black-Litterman Equilibrium.

Uses BL-implied equilibrium returns derived from the current SAA weights as the market portfolio proxy: $\pi_{eq} = \delta \cdot \Sigma \cdot w_{SAA}$ with risk aversion $\delta = 2.5$. These are the returns that would make the SAA weights mean-variance optimal under the parameterised risk aversion. The result is an internally consistent set of expected returns anchored to the current allocation.

16.2 CMA Output Tables

The **CMARetVolTable** shows one row per asset with columns: Asset, ETF ticker, Exp_Return_Ann (annualised expected return, editable), Vol_Ann (annualised volatility, editable), and Sharpe (ratio computed from the editable fields). Expected return and volatility are manually editable — enabling the analyst to overlay subjective views on top of the quantitative estimate.

The **CMACorrTable** shows the full correlation matrix derived from the historical return data (non-editable). The combination of analyst-specified returns and volatilities from the CMA table with historical correlations forms the complete set of optimiser inputs when **Use CMA as Optimizer Inputs** is pressed.

17. Data Tab

The Data tab shows the raw loaded dataset as a scrollable table. When prices are loaded, the table shows the raw price series. When returns are computed (from either prices or direct return import), the table shows the return matrix. This tab serves as an audit view — allowing the user to verify that data was parsed correctly, identify any obvious outliers or missing values, and confirm asset names and the number of observations loaded. The table is read-only.

18. Export & Report Generation

18.1 Export Weights CSV

The **Export Weights CSV** button on the left panel writes the current `app.W` weight vector to a two-column CSV file (Asset, Weight). The file is saved to a user-specified location via a standard save dialog. The exported weights reflect the combined SAA+TAA allocation as currently displayed in the Combined tab.

18.2 Generate PDF Report

The **Generate PDF Report** button (highlighted amber in the left panel) produces a structured PDF summary of the current portfolio state. The report includes: portfolio metadata (project name, date, configuration), Summary tab metrics, the weights table, the SAA/TAA allocation tables, the tracking error summary, and key charts (distribution histogram, sensitivity, efficient frontier if computed). The report is timestamped and saved to a user-specified location.

18.3 Save / Open Project

The **Save Project** and **Open Project** buttons on the left panel serialise and restore the complete application state using MATLAB's `save`/`load` with a `.mat` container. The saved state includes:

- `app.R` — full T×N return matrix
- `app.AssetNames, app.Dates` — asset registry and time index
- `app.W` — current combined weight vector
- `app.SAA, app.TAA` — allocation tables
- `app.Measures` — loaded macro measures for bootstrap
- All settings: Freq, DataType, Alpha, Horizon, RF, LongOnly, LB, UB, Objective, Optimizer, TEMethod, RandSeed

19. Internal State Variables Reference

All internal state is held in private properties of the `PortfolioRiskOptimizerApp` class. These variables are the single source of truth for all computation. The table below documents every state variable.

Property	Type	Default	Description
R	double T×N	[]	Returns matrix. T observations, N assets. All analytics operate on this.
Dates	datetime or numeric T×1	—	Time index aligned to rows of R.
AssetNames	cell 1×N	{}	Asset names matching columns of R.
W	double N×1	[]	Combined (SAA+TAA) weight vector. Updated by Optimize or Apply TAA.
Freq	char	'Daily'	Return frequency: 'Daily' (k=252), 'Weekly' (k=52), 'Monthly' (k=12).
DataType	char	'Prices'	'Prices' (auto-convert to returns) or 'Returns' (use as-is).
ProjectName	char	'Untitled Portfolio'	Label shown in report headers.
Alpha	double	0.95	VaR confidence level. Range [0.5, 0.999).
Horizon	double	1	VaR horizon in periods. Range [1, 252].
RF	double	0.00	Risk-free rate per period (e.g., 0.0001 ≈ 2.5% per year for daily).
LongOnly	logical	true	Enforce $w_i \geq 0$ constraint in optimisation.
LB / UB	double	0 / 1	Per-asset lower/upper weight bounds (same bound for all assets).
Optimizer	char	'auto'	Solver selection: 'auto', 'quadprog', 'fmincon', 'ga'.
Objective	char	'Max Sharpe'	Active objective function.
TargetRet	double	NaN	Target return or target vol for constrained objectives.
SAA	table	table()	SAA weight table: Asset, WeightPct, Description.
TAA	table	table()	TAA delta table: stored in TAATable.Data (10 columns).

Property	Type	Default	Description
NormalizeCombined	logical	true	Auto-normalise combined funded weights to 100% on Apply TAA.
TEMethod	char	'Var-Covar'	TE covariance method: 'Var-Covar', 'Historical', 'Exp Weighted (λ=0.94)'.
Measures	table	table()	Macro measures for bootstrap conditioning (date-indexed).
LastBootstrapDist	double	[]	Stored bootstrap path-return distribution from last run.
RandSeed	double	42	Random seed for Monte Carlo VaR reproducibility.
MCPaths	double	20000	Number of Monte Carlo paths for VaR simulation.
FredApiKey	char	''	FRED API key, read from environment variable FRED_API_KEY on startup.
AssetETFMap	cell Nx2	{}	Maps asset names to ETF tickers for Yahoo/Bloomberg download.
FactorBetas	double	[]	Last OLS factor beta estimates from updateFactorDecomp.
FactorNames	cell	{}	Factor labels corresponding to FactorBetas.

Appendix A: Tab Quick Reference

Tab	Key Contents	Main Action
Summary	Metrics text (mean, vol, Sharpe, Sortino, Calmar, MaxDD, VaR, CVaR@90/95/99, skewness, kurtosis, scenario lines), WeightsTable (Asset, Weight, RC)	Compute Risk
Distribution	Return histogram + KDE, VaR/CVaR markers with shaded tail	Compute Risk
Sensitivity	Asset shock chart (+1% per asset), Factor sensitivity chart	Compute Risk
Stress	Editable ScenTable (named shocks), pre-built episodes dropdown, scenario stats	Load All Stresses / manual edit
Data	Raw price/return matrix (read-only audit view)	—
SAA	Funded weight table (editable), FX overlay table, FX exposure summary, SAA analytics text, SAA stress table	Inline weight edit / Reset SAA
TAA	Asset selection list, TAATable (10 columns), TAATradeTable (trade-level summary), Apply TAA button	Add Selected / Add Trade Pair / Apply TAA
Combined	Funded table, Overlay table, analytics text (IR/Hit Rate/ETL/CDaR), TE method dropdown, TETable (Euler attribution)	Apply TAA (auto-refresh)
Perf	Cumulative return chart, rolling vol chart, drawdown chart, rolling metrics table	Compute Risk / window edit
Bootstrap	Condition builder (BootCondTable), block/paths config, bootstrap distribution chart, condition timeline, stats table, summary text	Run Bootstrap
Corr	Correlation heatmap, rolling correlation vs portfolio chart, PCA eigenvalue table	Compute Risk / window edit
Frontier	Efficient frontier chart (Vol vs Return), individual assets, current portfolio, distance-to-frontier label	Compute Frontier
RiskDecomp	Risk budget table (SAA/TAA split, MCR, CR in bps), waterfall chart, concentration metrics (HHI, ENB, DR), factor exposure table, rolling factor betas chart	Compute Risk
CMA	Method dropdown (Historical/Building-Block/BL-Equilibrium), lookback, CMA return/vol table (editable), correlation table, Use as Optimizer Inputs button	Compute CMA

Appendix B: Mathematical Notation

Symbol	Definition
R	T×N matrix of asset returns (rows = time, cols = assets)
w	N×1 portfolio weight vector; Σw_i = 1 for funded assets
w_SAA	N×1 SAA weight vector
w_TAA / w_combined	N×1 combined (SAA+TAA) weight vector; app.W
w_diff	N×1 active weight vector: w_TAA – w_SAA
μ	N×1 asset mean return vector: mean(R, 1)′
Σ	N×N sample covariance matrix: cov(R)
σ_p	Portfolio per-period volatility: √(w′Σw)
r_p	T×1 portfolio return series: R·w
k	Annualisation factor: 252 (daily), 52 (weekly), 12 (monthly)
r_f	Risk-free rate per period
α	VaR confidence level (default 0.95)
h	VaR horizon in periods (default 1)
VaR_α	Value at Risk: positive loss at confidence level α
CVaR_α / ES_α	Conditional VaR (Expected Shortfall): mean loss beyond VaR_α
MCR_i	Marginal contribution to risk of asset i: (Σw)_i / σ_p
RC_i	Component risk contribution of asset i: w_i · MCR_i
TE	Tracking Error vs SAA: √(w_diff′Σw_diff) · √k
π_eq	BL equilibrium implied return: δ·Σ·w_market
δ	Risk aversion coefficient (BL default = 2.5)
τ	BL confidence in CAPM prior (default = 0.05)
λ	EWMA decay factor (default = 0.94, RiskMetrics standard)
HHI	Herfindahl-Hirschman Index: Σ(share_i)^2 — concentration measure
ENB	Effective Number of Bets: 1/HHI(risk contributions)
DR	Diversification Ratio: (Σ|w_i|·σ_i) / σ_p

Appendix C: Recommended Workflow

Step 1 — Load data	Press Load Returns / Prices CSV, Use Sample Data, or Download Market Data. Verify in the Data tab that assets and observations loaded correctly.
Step 2 — Configure risk settings	Set VaR Method, Alpha, and Horizon in the left RISK section. Check Frequency matches your data.
Step 3 — Compute Risk	Press Compute Risk. Review Summary (metrics), Distribution (tail shape), Sensitivity (shock exposures), Corr (correlations), RiskDecomp (factor/asset contributions).
Step 4 — Define SAA	Go to the SAA tab. Edit WeightPct values to set your strategic benchmark. Add descriptive labels. Check the SAA Risk/Return Analytics section.
Step 5 — Add TAA overlays	Go to the TAA tab. Select assets and click Add Selected or Add Trade Pair. Set DeltaPct for each position. Press Apply TAA → Combined.
Step 6 — Review TE attribution	Go to the Combined tab. Check total TE vs SAA and per-asset breakdown. Try different TE methods (Var-Covar / Historical / EWMA) to stress-test the estimate.
Step 7 — Optimise (optional)	Set Objective and Constraints in the left panel. Press Optimize Weights. Review the frontier in the Frontier tab. Compare optimised vs current weights.
Step 8 — Stress test	In the Stress tab, Load All Stresses to see scenario P&L.; In the Bootstrap tab, configure macro conditions and run conditional bootstrap.
Step 9 — CMA (optional)	Go to the CMA tab. Compute expected returns using the preferred method. Edit the CMA table to add any analyst overlays. Press Use CMA as Optimizer Inputs, then re-run Optimize Weights.
Step 10 — Export	Press Export Weights CSV to save the weight vector. Press Generate PDF Report for a structured summary. Press Save Project to archive the full session.